

Sequences

Introduction: Patterns in Nature, Art, and Numbers

Everywhere people look, patterns emerge. The spiral of a seashell, the arrangement of sunflower seeds, the alternating colors of a tiled floor, the rhythm of a song — all follow a certain order. These patterns are not merely pleasing to the eye; they reveal underlying mathematical structures that can be described, predicted, and extended.

In mathematics, a **sequence** is a list of numbers arranged in a specific order according to a rule. Each number in the list is called a **term**. Sequences appear in countless real-life situations: the growth of a savings account, the depreciation of a car's value, the number of people in a growing city, and even the way a virus spreads through a population. Understanding sequences allows individuals to recognize patterns, make predictions, and solve problems across various fields.

This reading material examines the definition of sequences, the distinction between finite and infinite sequences, the classification of sequences (arithmetic, geometric, Fibonacci, and others), and the practical applications of sequences in daily life. It also explores the concept of **series** — the sum of the terms of a sequence — and how sequences and series work together to model real-world phenomena.

Part 1: What Is a Sequence?

Definition of a Sequence

A **sequence** is a list of numbers arranged in a specific order, following a rule or pattern. The numbers in the sequence are called **terms**, and each term has a specific position (first term, second term, third term, and so on).

Notation: The first term is denoted by a_1 , the second term by a_2 , the third term by a_3 , and in general, the n th term is denoted by a_n .

Examples of Sequences

Sequence	Rule
2, 4, 6, 8, 10, ...	Add 2 to each term to get the next term
3, 6, 12, 24, 48, ...	Multiply each term by 2 to get the next term
1, 1, 2, 3, 5, 8, 13, ...	Each term is the sum of the two preceding terms (Fibonacci)
1, 4, 9, 16, 25, ...	Square numbers: $1^2, 2^2, 3^2, 4^2, 5^2, \dots$

Finite vs. Infinite Sequences

Type	Definition	Example
Finite Sequence	A sequence with a specific number of terms; it has a beginning and an end	2, 4, 6, 8, 10 (5 terms)
Infinite Sequence	A sequence that continues without end; it follows a rule forever	1, 2, 4, 8, 16, 32, ... (never stops)

Notation for infinite sequences: Ellipsis (...) is used to show that the sequence continues indefinitely.

Part 2: Types of Sequences

Arithmetic Sequence

Definition: A sequence in which each term after the first is obtained by adding a constant number (called the **common difference**) to the preceding term.

General form: $a, a+d, a+2d, a+3d, a+4d, \dots$

Formula for the nth term: $a_n = a_1 + (n-1)d$

Where	Meaning
a_n	The nth term
a_1	The first term
d	The common difference
n	The position of the term

Example: 3, 6, 9, 12, 15, ... (add 3 each time)

Geometric Sequence

Definition: A sequence in which each term after the first is obtained by multiplying the preceding term by a constant number (called the **common ratio**).

General form: $a, ar, ar^2, ar^3, ar^4, \dots$

Formula for the nth term: $a_n = a_1 \cdot r^{n-1}$

Where	Meaning
a_n	The nth term

Where	Meaning
a_1	The first term
r	The common ratio
n	The position of the term

Example: 2, 6, 18, 54, 162, ... (multiply by 3 each time)

Fibonacci Sequence

Definition: A sequence where each term after the first two is the sum of the two preceding terms.

General form: $a_1, a_2, a_1+a_2, a_1+2a_2, 2a_1+3a_2, \dots$

Standard Fibonacci sequence: 1, 2, 3, 5, 8, 13, 21, 34, 55, ...

Rule: $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$

Examples in nature: The Fibonacci sequence appears in the arrangement of leaves on a stem, the petals of flowers, the spirals of pinecones, and the branching of trees.

Other Types of Sequences

Type	Example	Rule
Square numbers	1, 4, 9, 16, 25, ...	$a_n = n^2$
Cube numbers	1, 8, 27, 64, 125, ...	$a_n = n^3$
Triangular numbers	1, 3, 6, 10, 15, 21, ...	$a_n = n(n+1) / 2$
Fractional sequences	1/2, 1/3, 1/4, 1/5, ...	$a_n = 1 / n+1$

Part 3: Finding the nth Term of a Sequence

To find any term in a sequence, the rule must first be identified.

Example 1: Arithmetic Sequence

Problem: Find the 10th term of the sequence: 5, 11, 17, 23, ...

Solution:

- First term $a_1=5$
- Common difference $d=11-5=6$
- Number of terms $n=10$

$$a_{10}=5+(10-1)\times 6=5+9\times 6=5+54=59$$

Example 2: Geometric Sequence

Problem: Find the 6th term of the sequence: 3, 6, 12, 24, ...

Solution:

- First term $a_1=3$
- Common ratio $r=6\div 3=2$
- Number of terms $n=6$

$$a_6=3\times 2_{6-1}=3\times 2_5=3\times 32=96$$

Example 3: Finding Missing Terms

Problem: Given the sequence 4, 8, 12, 16, __, __, find the next two terms.

Solution: The common difference is 4. The next terms are 20 and 24.

Rule: Add 4 each time.

Part 4: Series — The Sum of a Sequence

Definition of a Series

A **series** is the sum of the terms of a sequence. While a sequence is a list, a series is an addition.

Sequence	Series
3, 6, 9, 12	$3 + 6 + 9 + 12 = 30$
2, 4, 8, 16	$2 + 4 + 8 + 16 = 30$
1, 3, 5, 7, 9	$1 + 3 + 5 + 7 + 9 = 25$

Sum of an Arithmetic Sequence

The sum of an arithmetic series is given by:

$$S_n = n(a_1 + a_n)/2$$

Where:

S_n = sum of the first n terms

n = number of terms

a_1 = first term

a_n = the n th term

Finite vs. Infinite Series

Type	Definition	Example
Finite Series	The sum of a finite number of terms	$2 + 4 + 6 + 8$ (only 4 terms)
Infinite Series	The sum of an infinite number of terms (may or may not have a finite sum)	$1 + 1/2 + 1/4 + 1/8 + \dots = 2$

Sigma Notation

Sigma notation (Σ) is a shorthand way to write a series. The Greek letter Σ (sigma) means "sum."

General form:

$$\sum_{n=1}^k a_n = a_1 + a_2 + a_3 + \dots + a_k$$

Where	Meaning
$n=1$	The starting index (first term)
k	The ending index (last term)
a_n	The general term (expression in terms of n)

Example: $\sum_{n=1}^5 2n = 2(1) + 2(2) + 2(3) + 2(4) + 2(5) = 2 + 4 + 6 + 8 + 10 = 30$

Part 5: Partial Sums

The **partial sum** S_n is the sum of the first n terms of a sequence.

Notation: $S_n = a_1 + a_2 + a_3 + \dots + a_n$

Example: For the sequence 3, 7, 11, 15, 19, ...

- $S_3 = 3 + 7 + 11 = 21$
 - $S_4 = 3 + 7 + 11 + 15 = 36$
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Part 6: Real-Life Applications of Sequences

Scenario: A student saves ₱50 in the first week and increases savings by ₱15 each week.

Week	Amount Saved
1	₱50
2	₱65
3	₱80
4	₱95
5	₱110

Questions:

- How much will be saved in Week 10? $a_{10} = a_1 + (n-1)d = 50(10-1) \times 15 = 50 + 135 = \text{P}185$
- Total saved after 10 weeks: $S_{10} = n(a_1 + a_n)/2 = 10/2 \times (50 + 185) = 5 \times 235 = \text{P}1,175$

Bacterial Growth (Geometric Sequence)

Scenario: A bacterial culture doubles every hour. Initially, there are 100 bacteria.

Hour	Number of Bacteria
0	100
1	200
2	400
3	800
4	1,600

Question: How many bacteria after 6 hours?

$$a_6 = a_1 \cdot r^{n-1} = 100 \times 2^{6-1} = 100 \times 32 = 3,200 \text{ bacteria}$$

Population Growth (Geometric Sequence)

Scenario: A city's population increases by 5% each year. The current population is 1,000,000.

Question: What is the population after 10 years?

$$a_{10} = a_1 \cdot r^{n-1} = 1,000,000 \times (1.05)^9 \text{ (approximately 1,551,328)}$$

Part 7: Distinguishing Between Sequences and Series

Feature	Sequence	Series
Nature	A list	A sum
Notation	$a_1, a_2, a_3, \dots$	$a_1 + a_2 + a_3 + \dots$
Example	2, 4, 6, 8	$2 + 4 + 6 + 8 = 20$
Question it answers	"What are the terms?"	"What is the total?"

Key insight: Every series comes from a sequence, but not every sequence needs to be summed into a series.

Key Terms

Term	Definition
Sequence	A list of numbers arranged in a specific order following a rule
Term	Each individual number in a sequence
Finite Sequence	A sequence with a definite number of terms
Infinite Sequence	A sequence that continues without end
Arithmetic Sequence	A sequence where each term is obtained by adding a constant (common difference)
Geometric Sequence	A sequence where each term is obtained by multiplying by a constant (common ratio)
Fibonacci Sequence	A sequence where each term after the first two is the sum of the two preceding terms
Series	The sum of the terms of a sequence
Finite Series	The sum of a finite number of terms
Infinite Series	The sum of an infinite number of terms
Partial Sum (S_n)	The sum of the first n terms of a sequence
Sigma Notation (Σ)	A shorthand symbol used to express a series

Review Questions

Responses should be written on a separate sheet of paper.

1. Define a **sequence** in your own words. What is the difference between a finite sequence and an infinite sequence?
2. Identify the type of sequence (arithmetic, geometric, Fibonacci, or other) for each of the following:
 - 3, 9, 27, 81, ...
 - 10, 7, 4, 1, -2, ...
 - 1, 4, 9, 16, 25, ...
 - 1, 3, 4, 7, 11, 18, ...
3. Find the 8th term of the arithmetic sequence: 7, 12, 17, 22, ... Show your solution.
4. Find the 6th term of the geometric sequence: 4, 12, 36, 108, ... Show your solution.
5. A student saves ₱100 in the first week and increases her savings by ₱25 each week.
 - a. How much will she save in the 12th week?
 - b. What is the total amount saved after 12 weeks?